

The Physics of Sentience with Professor Karl Friston

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I'm just taking the back end of your uh published bio talking about fellow of the Royal Society biology 2012 uh Welden Memorial prize 2013 uh embo um uh Excellence priz 2014 Academia Europa 2015 thankfully you haven't updated since uh on 2016 Charles brnch award for for on on parallel contribution brain research and the glass brain award uh really glad to have you Among Us um and thank you over to you for the physics of sentience thank you very much for that wonderful introduction and your kind comments uh it's a great pleasure to be here and it does feel like a very warm Community um I'm going to share my screen and then hopefully that will set us all up so let me repeat my um my thanks uh for being here I'm going to start with um as usual with this talk an apology but a very specific apology now I don't have a foundational training in cybernetics um but I have over the course of the work um largely inspired by Neuroscience but appealing to physics during my career it has become very clear that cybernetics um and the kind of work that I've been engaged with have very intimate relationships and probably share certain foundations along with things like perceptual control theory so um but I repeat I do not have a foundational training um in these disciplines so what I've got to offer in the first hour of this session um is really a view of self-organization and sentient behavior from through the lens of a physicist and neuroscientist in the fond hope that you will be able to join the dots and see how it can be translated in terms of uh your the the basic principles which with which you will be much more familiar um so I'm going to divide the talk into uh three parts first of all um telling a story about Behavior about um self-organization from the point of view of statistical physics and information Theory um with a special focus on markof blankets and how markof blankets individuate things that possess attributes such as behavior um and trying to understand those attributes in terms of something that we refer to as a Bas in mechanics um and then I'm going to tell the exactly the same story but using the concepts and the constructs that come from neuroscience science of neurobiology so how would these um physics like principles be applied to the brain uh to things like you and me um or indeed single cells uh and I'm going to illustrate that in terms of

predictive coding and neuronal networks as a particular instance or application of the physics that we're going to cover in the first part and then if we have time um I'd like to just um drill down on the nature of agency within this kind of treatment um and specifically um talk about the difference between mere reflexive behaviors of the kind you might say find in a thermostat um and the sort of planning um and deliberate behavior and agency that we find in things like uh things like ourselves um and I would you know I'm looking forward to a conversation about whether that bright line is necessary and how it does relate to um um cybernetics I'm going to start with um a question posed by scoder how can the events in space and time which take place within the spatial boundary of a living organism be accounted for by physics and chemistry I'm not going to answer that question and clearly we can't do that um but what we can do is just um highlight the notion of a spatial boundary as being requisite to be able to talk about anything in the sense that to talk about some entity some particle some person or something you need to be able to individuate it from everything else in the universe so I'm going to read that thing as a statistical object or the boundary that defines is definitive of that thing as a statistical boundary um specifically a markof blanket um so for those of you who are not familiar with the mark of blanket um notion it's a very ubiquitous construct in uh statistics SS and in causal inference um but plays a foundational role in what I'm about to say I repeat simply because it is the device that allows you to separate or individuate um the internal States and the blanket states of something from everything else so more specifically I've just tried to cartoon the intuition here in terms of a very simple little Universe where these blue circles represent some abstract state of the universe and the arrows denote an influence of one state on another state and if I were to pick um a particular set of states and to say my States and we'll call my these states internal States or states that are internal to me the mark of blanket is defined in terms of the parents the children and the parents of the children of the internal States and the role of the Markoff blanket is to provide a kind of insulation a kind of separation between the internal States and the remaining States the external States here so technically what this means is that if I wanted to predict the Dynamics or evolution of my internal States given the rest of the Universe I would only need to know the markof blanket all the information about the entire universe that is pertinent to me to my internal States is contained upon this boundary or these Mark or this Markoff blanket um mathematically that just means that the internal states are conditionally independent of the external States given the blanket States so that's my definition of thingness um uh and I'm going to make a further move here I'm going to um make a bipartition of the blanket States into sensory States and active

States U according to the very simple rule that the um the sensory States influence but are not influenced by the internal States and similarly the active States influence but are not influenced by the external States um so I've now effectively got a partition of all the states in the universe or this particular little Universe um that enables me to identify a particle or some entity that comprises the internal States and the blanket states that themselves can be divided into active and sensory States and you may be asking why have I done this well it it organizes a way of thinking about all the kinds of systems that we um as scientists and philosophers like to think about two of my favorite systems are here uh the brain um so under this kind of partition defined by these sparse couplings um the arrows how can we cons consider the brain well the brain would have internal States all my um you synaptic activities my connectivity everything I need to list in order to define the state of the brain at this point in time and the internal states are going to influence the active States they my actuators my autonomic reflexes um that perform actions on the external states that then change my sensory epithelium my sensory states that are uh then going to influence my internal brain States and so the cycle begins and exactly the same causal dependency structure or separation or partition of States can be found um any way you look so for example a single cell will have intracellular internal states that are internal to say the actin filaments of a cell uh that are themselves beneath the sensory states that are pushed out into the external states that reciprocate in terms of causal influences uh by activating cell surface receptors that change the intracellular states that uh change the action and so forth on both uh both of these examples you have this notion that the inside is influencing the outside vicariously through the active States while the outside is influencing the inside again vicariously through the sensory states and this sort of completes a a circular causality that as a neuroscientist one could read as your a very elementary form of an action perception cycle um what I'm going to do now though is just to ask you to forget about the markof blanket we're going to now rehearse 101 physics um to the extent that we need to to pursue the argument having established the core physics uh at hand with going to put the mark of blanket back into play and see what special properties emerge um and I'm framing it like that because the starting point for this is exactly the same starting point that you would find in all kinds of mechanics whether it's quantum mechanics statistical mechanics or classical mechanics it starts really with uh an expression of the Dynamics of some universe and I've sketched that out here in terms of uh in this instance a lan equation so the rate of change of some states is just a function of the state um plus some random fluctuations Ω and I've cartooned that here um in terms of two states um uh and this can be read at many

many different scales uh for example it could be neuronal oscillations um in one cell of my brain as they course their trajectory of these two states this two states State space space over very f um time scales say appropriate for gamma oscillations uh it could be um my cardiac cycle going through the phases of the cardiac cycle it could be me getting up in the morning um having a cup of coffee doing my emails and working through the day could be an annual cycle the point being here we are interested in describing systems that have a particular characteristic in fact one could say um definitional uh definitional of them um being uh existing in the sense that they possess characteristic States um the characteristic in question is that they always the system always revisits States it was once in um and this could be read in many ways it could be read as possessing an attracting set or an attracting manifold on which the states evolve um it can be read as the solution to a non-e equilibrium steady state um and it can also be read um effectively instead of being tracing out particular trajectories we can read the density of these trajectories as a probability that you will find me in a particular state if you sampled me at random and that's the if you like the interpretation that I'm going to pursue so why I focusing on that interpretation well we know a lot about the Dynamics not of the states in and of themselves but of the dens distribution the probability distribution over um those States and um some of you may or may not be familiar with various expressions of the density Dynamics I WR it down here in terms of a poer plank equation so now it's a rate of change not of the states but of the probability density over the states that is expressed here in terms of the amplitude of the random fluctuations and the flow um and this equation is again ubiquitous in physics and you know you could read it as a continuous form of Master equation it could be the time independent scrutin equation wherever you look it's a fundamental equation that describes deterministically the evolution of the probability distribution but I've just said we're interested in describing systems that have a characteristic set of states such that this probability distribution doesn't change with time and that's the key move here so I'm basically looking for solutions for these density dynamics that are now characteristic of the Dynamics of systems of interest that possess an attracting set um and just by setting this to zero and rearranging it um specifically using in this instance the hels decomposition what I can do is Express the Dynamics the flow the way that I change my States over time purely as a function of the amplitude of these random fluctuations and the gradients of these probabilities um strictly speaking the log probabilities so and there's also another term here known as a solenoidal component so this is this is the equation I want to focus on um because it's got some really intriguing interpretations um before I sort of get into those interpretations uh maybe a good

idea to to explain why there are two terms here um describing these flows on these probability or log probability gradients um one U this decomposition basically is just describing the fact that you can always split the flow into two orthogonal components um the first one um say for example we think about water flowing down a bath um the plug hole in a bath um then the gradient flow is Flowing down the gradients whereas the solenoidal flow DED by Q here circulates on the iso probability Contours or um solenoidal shows this kind of solenoidal effect as it circulates around so it's just a way of breaking up the Dynamics into a gradient flow um on the one hand and some circular flow on the other hand um both of those have interesting implications um you know the gradient flow we're talking about is actually flowing if you like up log probability gradients it's like Gathering flowing up a concentration of the probability of finding in a particular State as if the system was trying to gather itself into its uh attracting set and resisting the dispersive effect of the random fluctuations the circular flow is interesting as well because it it says that the these kinds of systems must have a form of oscillation or cyclical aspect solenoidal aspect to them and of course if you think about biological systems they're full of biorhythms and oscillations and indeed you could almost say life cycles themselves are an articulation or an expression of this kind of um um ubiquitous kind of Dynamics so that's where we've arrived the solution to the density Dynamics has to have this functional form which can be expressed as a gradient flow on these log probabilities excuse me what we'll do now is uh bring the markof blankets back and see if there is another kind of mechanics that complements or U fits in um the classical or um statistical and and quantum mechanics that can be derived from these uh density Dynamics so that flow equation holds for all the states and the particular petition that we discussed um under the Markoff blanket so the internal States will be doing uh a gradient flow on the probability here of the uh the mark of blanket States uh as will the active States and notice by construction I can drop the external States because the internal States and the active states are not influenced by construction by the external States so this leads to a slightly simpler form of this um these gradient flows um that I'm going to read in terms of perception and action both in the service or look as if they're in the service of maximizing the log probability um of States so what are the the interpretations of this quantity this log probability of some sensory States given a Markoff blanket or given uh given given me um and I've um listed a few interpretations here um just to illustrate um the again the importance of this of this uh fundamental kind of Dynamics this gradient flow if you recall this probability scores the likelihood you will find me in these states there are states that part of my attracting set they are literally those states that attract me and in

that sense there are states that are valuable for me given my Markoff blanket u and all of these all that's that this equation is saying is it looks as if my internal States and the way I act upon my world or my external states are trying to maximize by flowing uphill u this quantity which can be read now as reinforcement learning if you're a control theorist it would be optimal control theory if you're an economist could read this in terms of expected utility Theory and that's nice because the negative u of this u value u if you're an information theorist will be known as the self-information u the implausibility of this sensory observation or outcome or event given u more simply surprisal or even more simply just a surprise u and from this we can u spin out u the principle of Maximum Mutual information or the infax principle equivalently the minimum redundancy principle and indeed the free energy principle u why the free energy principle well the free energy is just a bound u or proxy for this u this surprise and I'll unpack the functional form of this later on again trying to give it give an intuition as to how one can interpret the components of u the free energy equipping it with a you know a teleology or a functional interpretation which may or may not be necessary to simulate u the behaviors of of of Interest the average of self-information or surprise is just the entropy u which means that this u fundamental gradient flow will look as if it's trying to minimize the dispersional the entropy of the B the blanket States in this example the sensory States and of course that's just an expression of the Holy Grail u u physics u or the physics of self organization u as articulated Say by Herman's synergetics and if I was a physiologist it would just be a statement of homeostasis it would just be a statement that systems that persist in a physiological sense u are those that show a generalized homeostasis they are keeping their essential variables as it were within viable bounds that are characteristic of the physiology or the the kind of thing u that this system is there's a final interpretation that I want to pursue here which is the interpretation that a statistician would bring to the table u so if a statistician saw this expression she would think ah you are now writing down the probability of some sensory observation some data some sample u from the world or generated by some u process or world u given not me on my Markoff blanket but me considered as a model of u the way in which my sensory observations were generated and this would be known in statistics as u u the model evidence u also known as the marginal likelihood u of these sensory data and I emphasize the model of course because what we're saying here is that u to be u to solicit valuable outcomes and to minimize my surprise and to self-organize in a way that resists the second law if Second Law applied to open systems of this sort is just to be a good model of u my world and of course this is just u a statement of

the good regulator theorem from Ross Ashby and colleagues uh but I'm going to pursue exactly the same notion from the point of view of statistician um leading to things like the B-Bas in brain hypothesis um um also formulated in terms of evidence accumulation Gathering evidence for my model of the world um and predictive coding which is the example that I'll show later on that leads to a specific process Theory um that many people in the neurosciences uh subscribe to In terms trying to understand the Dynamics um of a brain doing its sense making and Prosecuting uh its actions upon the world so that's the um that's the physics part um and now we're going to tell the same story from the point of view as if I was lecturing to um people in Psychology or cognitive Neuroscience um but it's the same it's exactly the same same same ideas um I think um nicely illustrated by this um uh 16th century oil painter famed for painting still lifes that when viewed from a different perspective give you very different visual impression so if previously you saw a bowl of fruit and now you see a face the point he's making here is that you made that face this is a very active construction an interpretation a hypothesis an explanation a fantasy that you have used to explain this particular of uh set of Impressions on the sensory sector of your marov blanket um so just to unpack that a little bit more it really speaks to the brain literally as a purveyor of fantasies or literally a fantastic organ that is evincing very much an inside out kind of um um perspective on perception which somewhat contrasts with early 20th century view of a sort of um outside in where the brain was um so particularly say visual High hierarchies were thought to extract information uh from um the sensory input this turns that on its head will have done then it strange inversions and say no no um it it feels it is a it is a um um a better way of understanding perception um and sense making is to treat the uh to treat the problem as an active problem of construction and the active nature will become even more apparent um as we get uh deeper into the notion of active inference so this view um I sort of mentioned 20th century um sort of outside in perspectives but in fact they were predated but by um this inside out uh perspective by uh people like Herman V hel Holz and you could argue right the way back through can to Plato but beautifully summarized here by helz um as follows objects are always imagined as being being present in the fal division as would have to be there in order to produce the same impression on the nervous mechanism so again he's referring to this notion of imagining something on the inside and then predicting what you would see and then using what you're actually seeing to update revise your hypothesis U until you've got a good account of what's going on and he referred to this um in terms of um unconscious inference or or a perceptual inference very closely related to the Notions of Richard Gregory in um psychology um who

saw perception as hypothesis testing again emphasizing emphasizing an inactive aspect um to um to our perception sort of visually palpating the visual scene with our eyes to gather the right kind of evidence to test the hypothesis that this particular um set of visual inputs U was caused by this or that and these ideas have been taken up in machine learning um and artificial intelligence research to great effect um for example people uh like Jeffrey Hinton and Peter Dineen actually built a Helmholtz machine um as a mathematical metaphor image of the basing brain uh borrowing explicitly from um Bayesian statistics and the work of Richard Sutton uh which is where the free energy comes from so uh Fineman um if you like solved the problem of evaluating that marginal likelihood or model evidence um which is an intractable Problem by converting it into an optimization Problem by producing a bound upon uh the log marginal likelihood or that self information that we're talking about earlier on um in the context of um QED or Quantum electrodynamics so putting these two together they invented the Helmholtz machine um which is formally almost exactly what we would predict from the point of view of this physics understanding of sense making um and we want to now extend the story to include action so let's come back to this notion Helmholtz's notion of Impressions on the nervous mechanism so if that Bayesian interpretation of self-organization under a given attracting set that possesses a Markov blanket what would that mean or how could one describe sense making on the inside on the internal States well it would look as if me on the inside I am in receipt of these sensory Impressions on my sensory epithelia my sensory Veil say my retina um or my skin or my um my hearing apparatus and they would present themselves as Shadows um sparsely sampled Impressions upon my sensory Veil my the sensory sector of my Markov blanket and it would look as if my internal Dynamics would try to uh predict or explain what caused those um um those sensory Impressions um so how could we think of that in terms of the internal Dynamics well we already know how uh how to describe the internal Dynamics say my brain Dynamics my neuronal Dynamics because it must be an instance of one of these um fundamental flow equations um so I'm going to make a couple of moves here what I'm going to say is if I can read my internal States as parameterizing some variational or probabilistic representation or beliefs a probability distribution uh about the external States then I can interpret this um um fundamental gradient flow which I've now expressed not in terms of the log probabilities but in terms of the the free energies that stand in for the the self information or the uh log um the log values um I can now Express this in a way that somebody in engineering would immediately recognize and it would imagine also people and say perceptual control theory would immediately recognize the particular form here I've written

down as a kalman filter um or a simple form of um um bayesian filtering um where the solenoidal part would be read as the prediction so imagine that the internal States me now stood in for some expected states of the world so that the me was now an expectation of some say a gaussian probability distribution which means that if I had some expectation about the state of the world I could predict the rate of change of the state of the world now I don't know whether that's true or not but I can now use the gradient Flow Part the nons solenoidal or the dissipated part of the gradient flow um I can now interpret these gradients of free energy in terms of a prediction error and I can use that prediction error to update my prediction to give me a um the best guess about what's going on out there in terms of the external States so let me make that try and make that um uh even more intuitive um and just asked well what is this prediction error well I've just said mathematically it's just the gradients of the um the VAR variation free energy or the surprise or the self-information um but intuitively what is it well imagine I had this sensory impression on my eyes on my retina and I had an internal brain state that stood in for a belief that this particular set of sensory Impressions was caused by a howling dog now I've had a generative model that could generate what I would see if this hypothesis or expectation was correct I can then compare the ensuing prediction with the actual sensory Impressions to produce a prediction error and then what this equation is saying is I'm using that prediction error to drive changes in my internal States my neuronal activity say in order to eliminate the prediction to destroy the free energy gradient to attenuate and resolve the prediction error until I have found the best explanation for this particular uh pattern in terms of the prediction hence predicted coding um now notice here that this is just describing self-organization of internal States in relation to external States and um communicated via sensory States in terms of minimizing prediction error you'll never actually know what caused your sensation so in this particular example it was actually a dog as sorry a cat casting the shadow uh that looked very look much like a dog but it doesn't matter if I don't know all I need to do is to keep my prediction errors low as long as I live and then job done so that's a very sort of parsimonious and I find a nice way of just describing the imperatives to sense making uh um and action um in the sense we can forget about the physics and just reduce it all now to prediction error uh and you know again for those people in perceptual control theory this this should uh be a story um well uh rehearsed um so what about this sort of action perception cycle well there are two ways I can minimize prediction error I can either change my mind literally by changing my neural activity to change the predictions of my sensory input and make them closer to the sensations to resolve the prediction error in the that

we've just described but there's another way I can do that I can actively resample my sensorium to make my Sensations more like my predictions and this would be action changing Sensations to make them more like predictions as opposed to perception uh namely changing predictions to make them more like Sensations so humanistically what this means is that we would expect to see action fulfilling the prophecies or the predictions afforded by perception um in a very very simple way and just to illustrate the Simplicity now going to um go um sort of talk explicitly about predictive coding uh and particularly hierarch predictive coding of the sort that people currently in the neurosciences think is a nice way to think about message passing and Dynamics amongst different neural populations um in the brain um so um what I've done here is U draw a schematic of the visual system um in its entirety so starting off say with visual signals from the eye reporting visual input they come into a a nucleus subcortical nucleus called the lateral geniculate uh nucleus um and these visual inputs are then um in receipt of top- down predictions and the difference then is scored by a prediction error that then is sent to drive or update beliefs expectations represented by units encoding what is being predicted to try and eliminate this first order prediction error in a hierarchical context though we can just um repeat this process these are now um first order predictions that themselves are being predicted by second order predictions that are you used to compare with the first order predictions to produce a second order prediction error that then is used to drive these higher order predictions and then so on recursively to any uh hierarchal level desired so that would be a picture of hierarchal predictive coding or hierarchal message passing or basing belief updating all all words for the same thing that will be consistent with this physics view of self-organization what about the action well let's consider another another kind of input the kind of input that um in neurobiology would call proceptive input this basically just reports the state of my actuators the state of my motor plant might say my muscles um uh and this would come in say to the pontin nuclei in the brain and it could be in receipt of top down predictions about um where I feel my eyes pointing um and I could use the ensuing prediction error to revise my beliefs about where I'm looking but there's a much simpler way of resolving these proceptive prediction errors I can actually just return them to the external world and cause them to contract my muscles until they're reporting exactly what I predicted and of course what I've just described is this a classical reflex arc um sometimes described um in the 1980s and 90s in terms of the equilibrium point hypothesis the notion now is that my predictions are now Furnishing set points that are reflexively realized in the periphery by using the prediction errors to destroy themselves or the the implicit free energy gradients simply by acting upon the

external States this instance the muscles so that I move my eyes so I get a different kind of information um now notice here this is very much sort of a Clos Loop thing um but it is deeply informed by the predictions that are gathering evidence from all different modalities so these predictions are realized in a relatively close form um setting but the predictions in and of themselves are U inherit from some very deep hierarchical um sense making if you like under this predictive coding view so that's the that's the basic story I just want to um um end now with with a little bit more of a technical Nuance on this free energy um and a conceptual proposition um that would differentiate between the kind of reflexive behavior that I've just described and the sorts of behavior that you would um study in Psychology your decision-making under and certainty planning to do this um versus that um and just to sort of set the scene um I've just Prov another schematic here that summarizes the the generic computational architecture that underwrites this action perception cycle so the idea is got our Markoff blanket our internal States and our external States we have the sensory sector here providing the inputs that enable the um the neuronal Dynamics that are performing gradient flows on the um on the variational free energy uh that look as if they are optimizing beliefs maximizing the um the um the quality of representations or beliefs about the causes of these ex sensory States the external States here to form um a expectations about states of Affairs out there Beyond sensory um beyond my markof blanket and then these beliefs are used to generate uh particular predictions specifically of um my motor plant or my appropriate the the appar that I used to act upon the world um um uh and we can read these in terms of proception prediction eras that then drive action that changes external States then Supply New sensory States and so we have our action perception cycle now characterized or described in terms of perceptual inference and the ensuing action selection that is then driven by the predictions of what the motor plant um should um should be registering or reporting um via Pro perception you can get quite a long way with this um with this construct um in terms of producing btic behavior um this is one of my favorite examples um illustrates a few subtle points first of all um sometimes the structure that you ascribe to the world actually comes from your own head even though you think you are trying to just um guess the causes of your own Sensations um under this view or under this description um so what we've done here is equip a a synthetic agent with a generative model in which the agent thinks that there's some autonomous Dynamics um in this instance a heteroclinic cycle that comes from a a lot voltera um um um attractor um that just C Ates with the heteroclinic orbit here um and the generative model is configured such that the agent thinks that this abstract movement maps to movement in some extrapersonal ukian space and

it's pulling an invisible string uh spring my apologies um that is connected to her finger so she expects to see and feel a finger being pulled around in this iant orbit um and of course those expectations now become descending predictions of what she expects to feel and what she expects to see but because the reflexes of fulfilling the predictions about what she expects to feel her arm will actually or finger will actually move thereby generating the visual Impressions that she predicted and satisfying those predictions and what that looks like from the outside is that the agent is doing a very ele mental kind of handwriting generating the sensations that were predicted in both the prop perceptive and the uh the visual domain so just to summarize that we have these top- down predictions in the proceptive domain generating action uh and what could be called corer discharge um um generating visual predictions that are then fulfilled by self authoring um the causes of Sensations through actually acting and generating what I expected to see um one nice thing about this is we can actually suppress the proceptive prediction errors so there's no information that this agent um has available to it um that would suggest that she or it is actually moving um which would be a little bit like um seeing stuff that's caused by somebody else and then uh and by implementing this um this adjustment we can now um simulate action observation observing somebody else act uh uh and compare it to the simulated neurodynamics during action itself but from the point of view of the agent very little has changed she's still got all the Machinery to predict this kind of visual input um so can use exactly the same um predictive generative model generating the predictions the same neuronal Dynamics to actually explain the visual input and indeed if one look looks at the activity um of the simulated neuronal responses one can reproduce things you find in your B like Place field activity but crucially the same pattern of selective responses are listed during action and the observation of another performing the same action so that's uh that particular illustration which I repeat is is is not um unbi mtic um rests upon the mark of blanket structure that I described before um here detailed in terms of the dependences what I want to do now is just make one very small Moon that could have a profound effect on the way that one might describe um uh deliberative Behavior what I'm going to do is I'm going to remove the influence of the active states on the internal States my motivation for doing this is that if things get bigger and bigger and bigger and bigger at some point the internal States will lose the direct connection from the active States um and so I'm going to remove that and that has a very interesting consequence it means that from the point of view of the internal States the active States now become a vicarious cause of the sensory states which means that it look as if the brain is now encoding the internal states are encoding the active States as if there's a

distinction between my physical realized actions that I can only infer through the sensory consequences of actions I've just cartooned the fundamental distinction there from the point of view of this basing mechanics or interpretation of internal Dynamics in um um in terms of this self-evidencing or basic mechanics or gradient flow on the log model evidence um um I'm drawing a distinction between these um the behavior of large particles where the active States um no longer have direct access to the internal States um and I've drawn that in terms of the you know the presumed beliefs of simple things like cells where they don't have to represent their own action they you know the action is directly informing and influencing the internal Dynamics uh in contradistinction things like you and me um May well be better described as having beliefs about our own action and that's crucial and I mean basic beliefs not propositional beliefs here um and that's crucial because that takes us into a different world it takes us into a world um of planning and inference where now because I've got effectively beliefs about what I am doing I now have to infer what am I doing if I'm making an inference I have to have prior beliefs about um what the kinds of things that I do and the argument here is well what kind of things do I do well I'm a free energy minimizing thing therefore I must uh more I will more likely commit to those actions that minimize the free energy I would expect consequent upon that action and mathematically that leads to uh this is what the the last slide now um a really interesting um decomposition of um the terms that constitute these if you like generalized prediction errors um or the free energy bound on on the log evidence so forgive the equations but the reason I put them up is I I just want to see what happens to various interpretations of this um free energy functional or function of um sensory states and active States um when we move into the future when we now um appeal to the free energy as a way of evaluating the likelihood that I will act in this way or or that way um so what I've done here is is write out the full expression for the variational free energy um and I've written it in two different ways this is the kind of expression that somebody from statistics would be comfortable with and they would read this as complexity um minus accuracy where the accuracy is just a log probability of some sensory States given my belief about their causes the the external States here the complexity is interesting it's effectively um a Divergence um technically a Kullback-Leibler Divergence but can be read just as a difference between two kinds of probabilistic beliefs namely um the um what I believe um my um beliefs about states of the world um uh given my exposure um and the sensory information namely my posterior beliefs after seeing some uh data under some action um and the PRI beliefs so it's it's it really scores the degree to which I change my mind in receipt of my sensory Impressions um it's the

amount of Information Gain um or the cost I have to pay in order to move my PR beliefs my posterior beliefs as a result of this updating process driven by these prediction errors if you're in machine learning um you would rearrange these terms um so that you would separate the log evidence um that we've been talking about um um um from another K Divergence between my approximate posterior beliefs that come from the free energy and the exact posterior beliefs about external States given the sensory States um so this means because this can be never be less than zero this free energy is often called an Evidence bound um in machine learning um sometimes with the acronym elbo because use a negative free energy elbo and evidence lower bound the reason I take you through that is that if I now say well let's take these quantities and ask what would they look like in the future before I've actually got any Sensations so now let's take um the probability distribution over the sensations given I'm acting in this way and use them to take the average of these quantities and what happens is they the complexity becomes risk the accuracy or the inaccuracy becomes ambiguity and the Divergence and the log evidences become intrinsic and extrinsic value respectively so what are these quantities what would they look like how could one interpret these well if we just ignore the prior preferences the attracting States the uh the log evidence for a moment and assume that I am equally attracted to all states of Affairs what am I left with I'm left with something called intrinsic value in robotics it's also called intrinsic motivation um in the visual search literature it's called Bas in Surprise um what is it it's very simple it's just the degree to which I change my mind about the external States given my sensory observations under this action in relation to the same beliefs before without seeing the sensory observations so this basically scores the information gain or the reduction of uncertainty about the external States it's um if you like the imperative for exploration it is sometimes known as an EP scoring the epistemic affordance of acting in this way as opposed to acting in that way um where the epistemic affordance just reflects the amount of information that I will gain if I do this as opposed to that um from the point of view information Theory it's just um a the mutual information between the causes of my sensation and the sensory consequences the external and sensory States respectively let me take um some uncertainty off the table and see what happens let's assume that I can see all the sensory States um and um therefore the external States become the sensory States and therefore there's no ambiguity what am I left with well risk so what is risk well it's just the difference between what I think will happen if I act like this and what a priori I prefer to happen uh my attracting external States or indeed uh sensory States so in engineering this would be known as KL uh control in economics it could be

regarded as risk sensitive control where we where we ignore the ambiguity and then finally if I remove all uncertainty so there's no uh reducible uncertainty in this environment i' come become completely familiar with it uh we're just left with the um the log evidence or the extrinsic value the expected value which we started with which is just scoring the expected uh um um log probability of uh me um ending up in my attracting set or sensory uh subset of those attracting sets so the final example is really just using these equations to simulate another kind of agent that now really thinks about the future and acts in a way not reflexively to minimize perceptive prediction errors but to minimize the expected free energy or the expected uh prediction areas in the future by resolving UNC certainty um and what we've done here is to simulate a very simple agent um whose Universe just comes along in three flavors uh all her Sensations are caused by an upright face a sideways face or an inverted face the um the Restriction here is that this agent can only see a very small part of the visual field so has to choose very carefully where to look and does so in a way that responds to this epistemic affordance this intrinsic value or um expected Information Gain and chooses the information Rich parts of uh of of the image um shown here in terms of Dynamics and what is actually sampled and in so doing resolves uncertainty about the three hypotheses um that best explain this sequentially sampled um um sensorium or visual input um showing a progressive reduction in the uncertainty in terms of the amazing confidence intervals um and correctly believing now that she is sampling from an upright face so that can be much more succinctly expressed um as always by helmz um and indeed he has done that here each Movement we make by which we alter the appearance of objects should be thought of as an experiment designed to test whether we've understood correctly the en relations of the phenomena before us that is their existence in definite spatial relations and so with that it only remains for me um to thank those people whose ideas I've been talking about and of course to thank you for your attention thank you very much indeed thank you Cole thank you so much um wonderful thank you um and um I think we're you'll have time for plenty of discussion and questions um if I just start please I can see some questions coming in the chat but if I just start um so I think we're going to talk some more about the good uh regulator theorem so no need for me to cover that off initially but let's just maybe explore one point and please don't um uh don't U make the mistake of thinking that um I I maybe all I need here is a repetition of something that you've all already said but when we decouple the internal States from the active um I'm trying to explore the real the way in which this is reliant on free energy minimization in other words this decoupling if I can or disconnection if I can call it that of the um internal in the sense is it

happening because of the principle of free energy minimization or in the sense is it required by that principle that's a really excellent and deep question and I did not cover it in uh in this presentation because um that the answer to that question um if correct has only um become apparent in the in the past few years um prior to much of the preparation of the material and the story that I was telling today um it has a clear answer it is not a consequence of the free energy principle uh I think generically speaking the free energy principle is a description of things that exist and you know um um with a careful definition of what you mean by existence but also a careful definition of um the particular um dependencies that underwrite your definition of existence which is markov blanket so all the free energy principle says is if you give me a um a random dynamical system with a particular sparse coupling that permits the um existence of a markov blanket that in turn permits me to identify something as persisting through time um then the free energy principle can be applied should you want to and you may ask well why would you want to um for fun sometimes and literally great fun simulating things um scientifically that that's you know the simulation aspect is very important um you know um being able to actually write down the Dynamics of a system that you think can be explained by this kind of gen model or that kind of G model is quite useful it also has applications as an observation model in phenotyping certain behaviors say in Psychiatry um but sorry back to the question so notice that the um the the dynamic DCS do not inherit from the free energy principle the free energy principle is a description of the Dynamics of this kind of thing so the answer now is to your question is well what kind of things um would um first of all show gradient flows that um are precisely the these Paths of least action and in particular the kinds of things where there's a disconnect between or a removal of the influence of the active states on the internal States the the first question is things that are not subject to lots of random fluctuations so we're talking about big things um things that are certainly bigger than you know a um a Quantum scale in which the random fluctuations average away and the bigger they get the fewer of the random fluctuations until you actually get right you get to the um uh the you know scale of heavenly bodies for example where there are virtually no random fluctuations those disputed gradient flows disappear and we're just left with the Divergence free solenoidal flow the rotation of the you know Heavenly Bodies the you know the the Earth around the Sun for example um but on route as we get bigger and bigger and bigger um there will be um a substantial portion and an increasing portion proportion of the internal states that can't see the active states of the um of the um the Markov blanket so the the supposition here is if you are big enough you may be describable as planning which says that there are small

things are unlikely to plan so you know very in a very simple minded way you wouldn't expect this planning as inference to be a feature or necessary to describe the behavior of a bacterium or a virus but you might expect it to apply to a mouse and you'd certainly expect it to apply to uh to you and me um just because we are bigger so you so the free energy is a method or a description that can be applied to things um and when applied to big things then then then you can certainly um um then you would be licensed to incorporate this planning as inference in into into the description or the application of the method is that the kind of wonderful very eloquent thank you so much and um I appreciate that so Jonathan shall we have a look at the chat please Yes sounds good and um what we'll do we'll take a couple of questions from chat and then we'll go to you David um so couple of questions in chat so first one's from Clement Vidal um Clement do you want to read out the the question you put in chat or I can do it on your behalf sure I can do it um so how many levels of hierarchies of predictions exist in the human brain and do they correspond to the number of brain layers um the there's a j I'm smiling because um Chris frit and I make a joke yes there is number it's six and then and then um and then somebody asked why six and said well because we said so so it's it's a bit of a joke um but it's also not a joke really um because again it it comes down to very very simple um arguments um about the size of the brain and um just appealing in a very coar grained way to um so re alization group what tends to happen as you move from very lowlevel sensory levels in these hierarchical structures which you can read literally as deep World models or deep generative models where the Deep just is an expression the fact you have this hierarchical structure which I should add depends upon sparsity of connections and markup blankets so you can't Define a hierarchy unless other than in in terms of which connections are not there and that is another instance of a Markoff blanket so each layer is a blank Markoff blanket for for all the um in a centripetal kind of hierarchy all the internal States um that are higher than that so in these deep architectures what tends to happen is you get um uh as characteristic of the renormalization group you get a slowing of the Dynamics um and that and that s tells you well how many times can you slow down Dynamics um in a brain that only lasts for say 5200 years uh it's usually about six so you start off with the cognitive moment which is about 250 milliseconds so you keep on increasing it by an order of magnitude say and there there's an upper bound on the number of times you can do that and suddenly you encroach upon the lifetime um the time you know the time scale over which this Mark of blanket actually exists so probably six is a number bound um not to be confused with um the fact that the the so what I'm talking about here are levels in a a deep um world model or

generative model that are deployed As you move through increasingly higher levels of O of your brain so so from V1 to V2 from V2 to V4 which about here um about here to V5 which would be about here yeah as you get sort of closer to to to um retal and uh frontal regions um the brain itself um can be um um can be en visited like a sort of um like a cabbage that's covered with all all the all the you know the neurop pill um that contains all the um nodes that that are connected the gray matter that um sits like a sort of a lair which is about 3 to 5 mm thick on top of this cabbage where the Cabbage itself comprises all all the connections sparse connectivity um called the white matter cu the they're marinated with little fatty uh wire sheets um that that that cortex itself has a layed structure so you be careful when you ask somebody like me about layers and levels because the levels we're talking about in terms of abst generating um uh and contextualizing predictions so as soon as you have this notion of hierarchal generative model that's generating predictions and you're saying that the deeper levels um click over more slowly in Universal or clock time you're basically saying that the deeper level provide a context for fast things at the lower level and the fast things at the lower level provide a context for even faster things at the level below that um so that that that um notion of a lay is really a level in a hierarchical uh in a deep hierarchical uh um architecture and that's very distinct from the layers of the cortex the layers of the cortex are really interesting because um when one drills down on the biological implementation of that predictive coding scheme that I Illustrated with predictions and prediction error being passed around it looks as if the prediction errors live in the top three layers and the predictions live in the bottom three layers and you could actually look at the wiring of the brain and start try to to make sense of um the hierarchical message passing that now has a lamin or L specificity in the context of cortical layers so it can be a bit confusing it's a fascinating area to get the you know the computation Anatomy implied by um the sparse coupling and the you know the predictive coding like interpretations great thank you right thank you um go to chance Mills next and then to you day we so chance again do you want me to read out your question or or would you would you like to read it out yourself um I I think I'm all right with reading it out myself thanks um so the American Society for cybernetics proposes you know this uh move towards ontogenetic resilience in light of the change of environmental factors at an unprecedented Pace loss of biodiversity changes in uh weather patterns things which kind of anchor our identity and connection to the world so my question is this in light of the uh free energy principle what is the Adaptive strategy for dealing with uh such a situation um that's you know a very challenging question uh I'm one would be tempted to answer that on two levels first of all you

know how um how would you describe adaptation under the free energy principle you could also answer it how would you use the free energy principle to try and ameliorate some of the challenges that are currently facing as in terms of um um overconnectivity and runaway uh Dynamics a failure of self-organization which is what would be predicted under this the fearing principle perhaps I'll just make that point clear because I think it's I feel quite passionately about that um the everything that I have said in terms of being able to um describe self-organization to an attracting set which I think from is just a description of self-sustaining systems that is a definition of sustainability at the level of of um of certainly of physics rests upon the sparsity of coupling um it rests upon the connections that are not there so if you go around globalizing and increasing connectivity you're destroying markup blankets if you destroy markup blankets you destroy um um self-organization you know technically from a dynamical system perspective you're inducing oscillated deaths everywhere and one could argue that that's what we what we are potentially confronted with at the moment in terms of overconnectivity and a destruction of all the sparse coupling um that preserves those delicate structures that have self-organized them over Millennia organize themselves o over Millennia so if he wanted to apply the free energy principle um to render a particular system be it an ecosystem be it a financial Market be it a um uh um meterological system if you could intervene on it um to improve its resilience what you are looking at is a way of uh preventing the destruction of sparse connectivity to to preclude um uh over connectivity uh to to maintain those boundaries and that segregation in the right kind of way um which means you know I guess ideologically um putting more regulation in place uh it means having very very clear geopolitical boundaries you know Mark of blankets you know in a um in a sustainable way um um finding mechanisms that would preclude um uh breaches of Markoff blankets um you know if I was a um an oncologist these breaches would be basically cancer you know cells growing Beyond their natural boundaries because they now become um um um poor Regulators or or models of their local cellular econ Niche um or you could you look at the your Wars in Ukraine for example as the equivalent kind of failure of a boundary or Mark of blanket at a geopolitical level um so that's how I would apply the free energy um as an academic how will you describe this I think it speaks to something we mentioned before which is separation temporal scales all that I was talking about during the presentation was focusing on um one Temple scale which was inferring states of the world states of Affairs out there but exactly the same um mechanics applies to um the parameters of a generative model which as a neurologist would be the brain connectivity for example if you're machine learning

it would be the connection strengths or the weights in say a Transformer model um and ALS
Al to the structure of um these uh you know these good models these these generative models
U by structure I mean you know the number of levels in a hierarchical structure the number of
components the um the particular structure that um endows it with a deep or hierarchical
structure all of these things have to be optimized in relation to the free energy that takes you
into the world of evolution um um in the a very simple sense that the free energy statistics is
used to do Bas in model selection but you can also read natural selection as Nature's way of
doing Bas in model selection selecting the right phenotype that's a good model of that
phenotypes lived World um so you've got these different different time scales in play so I think
adaptation of the time the kinds of time scales that your question Intimates would have to I
think appealed both to parametric learning um and structure learning um um uh both of which
can be expressed as free energy minimizing or evidence maximizing uh processes um and
you're in the face of of a changing environment um that in in turn speaks to interesting
differences between inference and learning where inference is updating your um your
embodied beliefs about states of Affairs hidden latent States in machine learning talk um
learning would be Bas in belief updating the parameters of Your World models and then um
structure learning or selection we'll be selecting the right kinds of structures each one um
nested within uh the other with a separation of time scales so you know for example I can't
make sense of my sensorium um unless I've learned the right kinds of contingencies of um um
and I can't learn the right kinds of contingencies Say ne during neurod development unless I
got the right brain structure so every one of these scales depends upon the you know all the
other ones and of course that uh that also holds in reverse that you know my brain wouldn't ex
exist unless I had you know a structure and of course I wouldn't have the wiring uh in order in
order to do the message passing to make to do perception uh so all of these different scales
depend uh depend upon depend upon each other is that the kind of answer you were looking
for or yes thank thank you very much Professor David we'll go to you next and after David
we'll go to Trevor Hilda okay uh well thank you uh very much for elucidating and giving me a
far better feel for perception information processing and uh moving towards action I don't
want to pretend to have understood and digested everything that uh you've said um completely
uh Professor kisten what I'm interested in is how far you see this form of analysis of moving
towards kind of dissolving the uh sensation uh body problem the hard problem the issue of
qualia or um raw field is information um prior to sensation is sensation a concept we have no
need of or does it emerge out of processing our processing um so to speak I I just wondered if

you could sort of help me around this area or whether philosophically you regarded it as a non question no I I'm not allowed to regard it as a non- question I have too many philosophy friends but because I don't do philosophy it's it's not a question I can help you with but I can get I can point you to the kinds of answers on the literature where people are really focusing on this um uh but it is a philosophical question um from my point of view um you the question really reduces to what what kinds specifically what structures um um specifically of the gened models would be necessary in order for um some artifact natural or otherwise to be sentient in the way that you're you're you're implying yeah um and you know what most people end up um um end up with is a number of different arguments you can take the agency argument um which I quite like you know to be uh to be um sentient is to be an agent so it's me actually Gathering my data and possibly experiencing my data or or not but before I can experience it I have to be me and therefore I have to be an agent so immediately you're talking about um um how would you define agency in you know as a physicist um and if you defined agency of A non-trivial Sort as the kind that inherits from um the uh representing the consequences of my own action that tells you a number of things immediately it tells you that certain things possibly viruses cannot be sentient in the way that an agent could be it also tells you that um things that have sentients if they are agents must have a generative model of the future um in the sense that in order for me to predict the consequences of my action to plan I have to have a ger a model of the future in the sense that the consequences not yet occurred which means there's a certain temporal thickness to these World models that you possibly wouldn't find in a virus or a thermostat but you might find in me um so that might be one bright line between the kinds of things that can be sentent and well to have an agentic S sentience um and then you move on to um other perspectives um some perspectives emphasize as you mentioned the body so I'm thinking about the work with people like Anil Seth here and Lisa Feldman Barrett they they say that well to be to have minimal selfhood is just the hypothesis that the best explanation for this Myriad of Sensations from my eyes and my body my skin um the best explanation for it is I am an embodied being and I am me and all of this um is the best explanation for what's going on so now selfhood itself becomes another um becomes another um hypothesis um which you have to ask why would you have that hypothesis well one advantage of having that hypothesis once you realize I am me um is you can now exert mental action over um which things you attend to that's quite a long story here but I'm just slipping in the key words that that some people think that it's not any agency that makes you sentent it's a particular kind of agency that's on the inside it's basically attending to different sectors of your

Markoff blankets um that basically allows you to select what information you are going to use to do your updating about myself and my plans so there are people um usually German philosophers who would focus on on mental action as necessary for uh for that kind kind of sentience and and and minimal selfhood and then you get into um even even more um contentious issues about um um self-awareness and having models of me in different states of mind you know is the hypothesis that I am a person and I am anxious um is that part of my generative model and what would that look like in terms of James Langan um sort of formulations of your of of emotions and their bodily Expressions so really fascinating answers a question uh and lots of fascinating answers but but there are lots of them so I think you just have to take your pick from from your favorite philosopher I think that's yeah so will turning an artificially intelligent system off ever be morally equivalent to Knocking somebody out yeah um if you if you could engineer such such a um a sentient yes I think so I mean that's a little bit um I don't like knocking people out because that's usually has more as in a psychiatrist neurologist this is you shouldn't really do that if you can avoid it but certain like putting some to sleep or unitizing them yeah yeah it would be it would be very much like that the argument now is will you ever get um um a btic intelligence um you know in an artifact I don't see why not provided that artifact is embodied so that you know you give it the opportunity to learn oh I am embodied and I am a thing um and I can plan um and yes in principle yeah it would it would be like switching you you switching you off when you went to sleep for example you thank you um Trevor do you want to go next and then after treasure thank you thanks very much thank thanks Professor Frist um I wonder if you could comment on the thousand brains theory about how the neocortex works yes I can yeah so Jeff um um um Hawkins the inventure of the pine Palm P pilot for those of you who can remember the Palm Pilot uh so um there are lots of similarities but some key differences between the kind of um canonical micro circuits that that are um on offer to if you like um explain one instance of predictive coding the similarities um inherit from sort of a a common interest in trying to get a btic formulation of um sense making and action so in fact Jeff Hawkins hosted uh a meeting it must be 30 years ago now when I was a young man where you know we started talking about these things shortly afterwards he worked with a gentleman called diip George who's still doing quite um exciting work I think at newor together they came up with something called the hierarchical temporal model the HTM and that shares exactly the same fundamental characteristics that we're talking about before in terms of the um the separation of temporal scales As you move deeper and deeper into generative models so there's a you there's a shared

commitment um not just to generative modeling as a view a constructive view of sense making and and action and pneumonics um uh but that also extends into the very very specific architectures that speak to a separation of Temple skills um which was celebrated in Jeff's early work the Thousand brains one um does you know it's gone off in its own particular direction and I think that the know the connection here would be more like um a mixture of experts perspective on generative models so I think a very plausible uh notion that um you could have um a whole bunch of generative models all competing to explain the lowest level of a sensory hierarchy um and each one will be fit for purpose for a different context and one way that they can if you like resolve uh infer the particular context in which they're operating is to compete to see who can explain this sensory input the most so in statistics this is called Bas in model averaging uh where you wait the predictions of each model in proportion to the free energy or the evidence log evidence sorry the evidence for each model I think what Jeff has in mind from my point of view would be there you know there are millions of these little mixtures of experts that are all competing so I think it fits very comfortably with the overall free energy principle probably less comfortably with um predictive coding of the of with specific architectures that will be informed by neurobiology um you know the Thousand brains formulation is is not is not quite um doesn't have quite same commitments as the normal canonical microcircuit formulation of predictive coding uh and the empirical evidence speaks to the canonical microcircuit formulation just because the chical microcircuit formulation was designed to account for the empirical evidence whereas Jeff didn't have to worry about that because he he you know he's just trying to find the right architectures but a lot of shared commitments some differences some nuances and I I'd be interested to see what it comes up with in the next 10 years it's great thank you very much you um we'll go to Margaret Heath next Margaret I see you've got a couple of questions in there so if you want to cover both of those that'll be brilliant thanks hello yes thank you Professor fiston um thank you for all your work it's really it's it's yummy um I have a question that is on on two levels really um the one is uh to do you've discussed in other lectures um this notion of temporal death and counteract short breath and typically theories of the imaginary or the imagination think of this as as disconnected from sensory information or the external and something that happens internal to the organism curiously I'm quite interested in Andy's extended mind stuff and the distributed cognition Collective cognition where you'd have ensembles of Agents doing stuff like this so that would change the formulations a little I was wondering if you could talk to us about those sorts of ideas yeah I mean by Coincidence of course the you know the notion of

counterfactual breadth um that is if you like another dimension to the temporal depth um came from a Neil Seth and Neil Seth has an office next to Andy as um you know Andy clar being the author of the extended mind and and designer environment so I think there's a lot uh you know they they clearly have their own views of things but there's also a lot of synergy between between the two you know Anil focusing more recently on embodiment and being a beast machine where whereas um and is I think you know generalizes the importance of situatedness and and embodiment um um so um so I think there's a very a very close connection between those perspectives and you can trace the legacy of these ideas I'm sure you have done you know from you know historical writings that um have emerged in terms of people like Andy Clark and Neil Seth and and the like in terms of the you know um the IM the counterfactual breadth being divorced from the sensorium I think that must be right in this because there's no full divorces if you allow for this vicarious sparse connectivity that just is the the message passing or the belief updating in a hierarch architecture we're talking about but um I think sort of just common sensical yes if you are planning what to do and you're planning in the future you're planning a roll out uh a trajectory a path into the future clearly there are no um observable outcomes from the future so it is purely counterfactual and um as soon as you entertain more than one path into the future you don't have to I mean thermostat doesn't you it just considers one little tiny trajectory into the future but if if we're now um you know in the planning as inference kind of um um active influence or predictive processing then then you're going to have more than one path you're going to evaluate that and of course the the number of paths that you evaluate is exactly the counterfactual depth that Anil um was referring to and it also introduces interesting semantics which philosophers um like which of course once you've got um more than one path to select from you induce a problem of selection now as a mathematician that's fine that's Bas in model selection I just I just select the one with the greatest evidence or the the least expected free energy um but of course if you're a philosopher you've now got the notion of free will because you've got the act of selecting amongst the counterfactuals uh or Not Free Will depending upon uh if it's prescribed by yeah yeah so yeah I think um yeah I think yes it has to be imaginal in the sense that it is imagining a future and you can de articulate that under simulation theory for example um um and in that sense it is divorced from the s from the sensory input but of course it is underwritten by you you by the sensory input because your your paths do depend upon where am I at the moment and that is conditioned upon sense making um so yes and sorry just to um then make the link to if you think about Levan's itive cognition work all right so and the mark of blankets and and then I

listened to a really fascinating Chris Fields with you uh on Psy philosophy Babble I think it was where we spoke about um that boundary condition the information processing at on the boundary and then the distinction between two Mark of blankets and how to unpack that when are there one mark of blanket and when do you have to compute them as separate entities and with separate agency yeah you're bringing in a whole other fascinating World here you people like Mike L and Chris Fields uh um and you know they would talk about things like distributed cognition and basil cognition and ascribing the more Elemental kind of active infant I was describing ascribing them to sort of even molecules or at least macro molecules and cells asking questions about the emergence of multi cellular organizations and the intracellular signaling and would this be described as well it certainly would and could be described as active inference and of course once you do that you can use the the teeology that you get from cognition and hence distrib cognition um so I I very much encourage that and um and encourage um Chris fields in his Mantra that you know in reality there should be no difference between um physics and psychology um and biology and philosophy they're all the same thing and we should all be using the same words quite merily and an interchangeable way and I would fully subscribe to that um the just one interesting example of that which um I I always like to reiterate um is you know once you understand that you or once you allow yourself to um with W with gay abandon use cognitive terms to describe the behavior of cells either maintaining or destroying their markof blankets you can now as Mike does and has done uh talk about cancer as a delusion you know a false inference and you can start to sort of think you know think in a way which I think is quite an expressive and interesting way to think about the know physical phenomena at the cellular or the B the molecular level in terms of this kind of um um um you know not ascribing intentions to to cancer but but there certainly the same kind of collective or um situated cognition that societies of PE of things like you and me uh you might might express it also the two final points here of course you know there isn't just one mark of blanket there are probably not an infinite number but certainly almost uncountable because of all the common toric so any sparsely coupled system you can draw a markup blankets everywhere um and of course you can have markup blankets of markup blankets so you really usually have to identify the markof blanket of interest and that defines you as a psychologist or molecular biologist or sociologist or ethologist or whatever uh that just you that's your scale of markof blanket and that kind of markof blanket is what you like to uh to talk about um so you know as we intimated before just having a deep generative model or generative model means there are multiple Markoff blankets in the brain and I

mentioned that because Chris more recent has coming back to to David's question he more recently um um uh tried to understand what you the locus of sentience in terms of Consciousness as um in terms of an inner screen um and or and the notion of irreducible Markoff blanket so there may be one Markoff blanket right in the middle of our brains in the pineal gland that cannot because of its lack of sparsity U contain any more Mark of blankets within it and that has some special characteristics that uh Chris argues might might U you know might be associated with Consciousness you introduced Fields there in terms of um his take on Mark of blankets his take is um beautiful and Abstract um uh and comes from Quantum information Theory where effectively he replaces a markof blanket with a holographic screen I think that you know that's that's what you were you were referring to which is a wonderful uh move uh sometimes a little bit beyond me because I've forgotten all my quantum physics but it is it is an interesting move to make um to think about the relationship between um communication and exchange between the internal and the external bulk um via a holographic screen because the holographic principal tells you all the stuff on the inside all the information on the inside bulk has to be on the surface on the blanket on the holographic screen which basically means that everything I need to know or that is knowable about you is on your Markoff blanket it's just how we exchange and that means I will never quite know what's going on inside your head and can never know by definition so that has some interesting uh consequence I think for it's just delightful isn't it thank you thank thank you um we'll go to Harris nei2 next and then Dave I see you put a number of comments in Dave so we'll then go to you after after Harris so Harris you could summarize the question you put in chat that would be great thank you yes great talk and really insightful answers uh thank you and here is my question uh scribes in ancient times were tasked with copying text by hand and it was a rather laborious and time consuming process uh those scribes aim to minimize the effort expended in the copy process much like physical and biological systems they strove to achieve the most efficient and accurate production of the original text while minimizing Surprise by reducing the likelihood of introducing errors uh they developed techniques uh such as using templates employing standard tise formats and even utilizing Pimon mneumonic devices uh improve efficiency and accuracy so Professor friston considering the parallels between the least action principle in physics and the free energy principle in cognitive science is it safe to utilize these principles to inform our understanding of human cognitive processes in activities that required uh this extra medul precision and optimization such as the man manuscript copying work of those cpes yes I I I think well is it safe to apply it um that's a

loaded question but a very nice question I think yes I think it would be perfectly viable to apply that and I think it's very good of you to actually introduce a notion of a you know a principle of least action uh because that's just what the free energy principle is it's nothing more than a principle of least action so you can articulate it using a path integral formulation literally as the um understanding both action and perception as evincing or just pursuing Paths of least action which is as you are intimating um Paths of Maximum efficiency Paths of minimum error Paths of Maximum replicability Paths of Maximum uh and and sustainable um um and resilient uh reliability um that that you know if you if if something was able to um persist and exist on its own attractor its own um in its own preferred States very very precisely you you would get exactly the behavior of your scribes so I imagine they had a very comfortable and relaxed life with this very very specific model of what they do and they did it perfectly um and you know and in that sense they I think would be an example of something that would easily be described in terms of a path of least action um and as such um you would be fair to apply the free energy principle to you know to to that to to that kind of behavior i g I guess what you'd have to ask though um is well if we are just trying to minimize prediction errors you know would you would your scribes ever become creative would they become poets would they start to experiment with different artistic Scrolls um uh and then one gets into the interesting questions well where does that kind of information seeking come from and then of course the answer from the point of view of the free energy principle it's only when you get big enough that you now have the opportunity to think about the consequences of your action and then that Curiosity that epistemic aordance that expected Information Gain starts to come into play so again you might think that scribes that committed their entire lives just to copying manuscripts were um were were U you know were not um probably got quite bored I assume they had quite big brains so so they probably wanted to to automate that and do something that sat in their curiosity uh because you can get a machine to do that because the machine doesn't plan in the it's not it doesn't have that epistemic explorative um capacity that that a machine does or you know uh um um a calculator would and I mean calculator in in the original sense of you know um people actually do calculations at the beginning of the Industrial Revolution so a very interesting question is that is that what you meant yes yes thank you so much great answer thank you thank you thank you after David we'll then go to arand van campen and then Richard Barry so David do you wantan to go next Dave Douglas me me great thanks okay welcome Carl okay I've been in many of your presentations at active inference Institute now today you've met mention the richness of the Markov blanket Concept

in various contexts now we know paper of yours from a few years ago I'm sorry I forget the title you mention that the blanket around an informational regime within uh a living tissue particularly neurological tissue might or might not coincide with cell membranes or tissue boundaries do you have any further thoughts you can share with us about that there are some people on on this side of the pond who would be very pleased to um reflect on your thinking about how one differentiates um a psychological process from the people in which that process embeds itself or a a meme from the various uh brain tissues that the meme uh embeds itself uh we we're updating the U work of Doc Professor Gordon Pasque who I think addressed some of these same questions in a very concise or incisive way and I think simplifies things by m simplifies the work to be done by slightly multiplying the number of conceptual tools to be used can can can you expand upon that last last sentence well I should repeat the the the the the gist the when we talk the the there's traditionally go going back to beralan there are there's been a lot of many many times people have pointed out that a lower or in your terms a faster level of activity may be controlled or steered or motivated by a higher level or a slower level of control so you have this res this asymmetric Mutual influence and control and communication Duality parel okay now usually the Assumption has been that there's a single way to perform that kind of analysis in fact that it's just a simple if you have multiple levels that if you have two more than two levels to deal with there is necessarily a strong or weak ordering among levels of control okay there's the word of God there's the traditions of men there's your physi there's your work as a farmer and there's your physiological needs and there's a strict hierarch or One A controls B controls C controls D and that that individuation occurs in one and only one way it's just control control is control presupposition is presupposition now Gordon P working in the 50s 60s and 70s along with some some of our colleagues still around still active in cybernetics discovered that well in the realm of teaching of of teaching and learning it's much simpler to say no wait a minute don't try to conflate all control relations into a single con there are after all psychological processes and there are mechanical or physiological processes the physiological processes are spatiotemporally distinguished M machines you can look at the boundaries they have a boundary in uh in space they have origins in time but the phys the psy and the psychological processes also have boundedness but it's not that same kind of boundedness where is Beethoven's Symphony I want to know where the exactly the spot temporal boundaries of that are it's it's a category here so Pas just says look there's individuation into Minds psychological entities memes and there's also individuation into spatiotemporally bounded things regions it is the case that mental

entities control and listen to other mental entities it's the case in um I'll say the organization of a thunderstorm perhaps or or you know dirty water is a better example turbulent dirty water that there's there's control of large uh small the the possibilities of small entities by large entities but the large entities are also dragged around by the small entities so you have that control and it Al also the case more importantly perhaps that sometimes a mental entity say a meme in the in the classic sense controls the the the um the web browsing activities of a number of individuals and at that point the psychologically differentiated individuals acting among themselves control and are influenced by physical entities the fact that your electricity went out today he knocks out an entire class of mechanical entities the computers aren't working and I know that in in one of your papers explaining the physiology of uh predictive processing and active inference you at the end of one of the sections in about one sentence say it's not clear that the boundaries of the that the functional processes always coincide with the boundaries between cells or between tissues that's the point where my ears stuck out and I said I remember Gordon pasque telling me about that in 1975 here's Carl frisen please tell us more about that is there more than one kind of individuation either instrumentally in sense in the sense that oh I feel like dividing the world up in this way or I can divide it up that way or I can divide it up both ways and then see how they fit together or maybe cells and memes and songs and cultures and legal traditions and biological species individuate themselves in various ways Dave I think I think you posed the question great Carl you want to respond yeah well there a lot to respond to that was Illuminating I didn't know about that work but it all sounds um very sensible from my point of view uh and the kind of work that I should know about um so in brief uh you know the way that you were talking about the separation temporal scales and the circular causality I understand that from um the work of say hman hen and you know in terms of you know the the the slaving principle and your top down and bottom up causation that and I think that that that is that that view is nicely evinced with the recursive message passing in these deep generative models it also makes entire sense that um you should be able to um apply things like the pre free energy principle to different kinds of things whether they are memes or communication or um um the products of um um products of culture for example right down to um amoeba and single cells that have spatial temporal pounes absolutely you know the free energy principle is just a principle which I mean by which I mean it is a method it's a tool that you can apply it to um could you apply it to um things with you know spacious temporal boundaries like a cell absolutely people do that uh could you apply it to something more abstract um such as the passing of memes on on the web for example absolutely people

have done that um and I think that you know the Markoff blanket there is not really quite so well defined spatial temporary it's more as you would Define it in a you know if you're a systems theorist it would just be input output relationships so you know in terms of systems theory the mark of plank is nothing more than defining the system in terms of its inputs and out puts so you'd have to sort of label the inputs and outputs on on your um um on your Bluetooth or your ethernet uh in a social web and that and that would be the you know the way of defining the mark the the conditional dep sorry the sparse connectivity that defines the Markoff blanket um and then it wouldn't quite Define Beethoven's Symphonies but it it would allow you to to to then talk about the Markoff blankets of this community or this social group and the exchange of ideas provided they could you know they could be classified or Quantified in some formal way as message passing between Markoff blankets and you know and you should be able to apply the free principle I say it's been done it's been done in a very Elemental way um by looking at um opulation voting Dynamics by simulating agents passing messages on the web but in a very simple way in proportion or in a way that that each agent believes in the veracity of another P agent's belief depending on how similar that agent is to itself um and then you get some really interesting polarization Dynamics through this multi-agent active inference that I think you could describe in terms of you know passing memes uh and the collective commitment to a meme I think would would start to speak to this more abstract mindful kind of application that that you were intimating yeah right thank you um we only got a few minutes left so i' like to squeeze in the last two questions so Arend do you want to go next and then we'll finish off with Richard yes thank you Dr friston really great to be here and see you live I I watched you on with Lex Ritman a while ago huh um my question would be uh I worked with Dr F of CA who I a picture I'm going to see here and and he says in 1991 so my question is would you agree with the observation of f of CA that Global disorder social entropy is caused by a crisis of perception um I think I will have to go away and think about that um it speaks very much doesn't it to the earlier question about you know could one use um free energy principle and related uh formulations to talk about um um um societal uh disorder uh in terms of communication and the way that we perceive others um I I I think it would be trivial of me to try and give you you know a clever or informed answer without thinking about this it it does uh I mean one useful thing uh I think that could be um brought to the table here is that the the free energy principle is dual to James's maximum entropy principle which of course is you know could be cast as what are the constraints on a particular kind of disorder but the twist um from the point of view of the free energy principle

is the entropy that James was talking about was the entropy of the measurement it was the observation and I think read as a perception I think there's a deep connection now between James's Maxim entropy principle which you know for him is how does say how does a physicist perceive by doing her measurements um and you know almost paradoxically saying that the best the best kind of perception is maximizing the entropy of the measurement representation which for for the free energy principle would be the um the variance of that um Q distribution over the external states that are being measured so I think there is some deep connection between perception as measurement and the and and observation and um the natural tendency of things to minimize their thermodynamic free energy and maximize their representational entropy um what that means for the entropy and the disorder of the of the thinging of themselves I'll have to think about yes I was just referring to the work of Kenneth Bailey a social entropy Theory uh which I've been looking at and that's why the perception the observation the interpretation of real reality differs right so that is the I think the the cause of many many misinterpretation and many Mis actions at this time causing this social entropy yes yeah we don't have time but you know it it would be really nice to talk about sort of um some of the pathologies of perception from the point of view of applying the fenry principle to psychiatric conditions like hallucinations and delusions sensory attenuation and what that would look like in terms of communities talking to each other making inferences about each other uh some really fascinating uh fascinating issues there yeah thank you sir thank you thank you Richard I'll you want to ask you do the last question and if everybody hasn't seen it there's a question to us all collectively from Paul pangara but we we won't have time to cover that so Richard you want to go ahead uh thank you thank you Professor fascinating to I'll try and um use pass and and minimize my uh use of of free energies um so um we make decisions as a society collectively um we draw inference from the world and information we are provided um sometimes very inaccurately um and I'll go back to the butler inquiry and the decision um to go to war on incorrect inference around weapons of mass destruction we see it daily in the courtrooms where um 12 good citizens will be asked to draw inference from um evidence that they perceive has been presented to them and make some some pretty high-risk decisions um around individuals and of course we get it wrong and the post post office inquiries is a really really good example of collectively getting it wrong so so my question really really straightforward um is there a way we can manage um what you've described in a a more consistent way and draw inference sorry have more confidence in the way we draw inference um from from world um I think the short answer um you would be um you're you're not going

to find a cure but you can certainly articulate the problem that would reveal strategies to cure and I you and I'm using the word cure presuming of course you don't want to have this um this this extremely unstable and unpleasant situation where you just don't know where to turn for for Reliable information so everybody has lost epistemic trust in everybody else with fake news and the like and I've also heard the argument just to generalize that um in terms of you know young people nowadays having so much access to different channels of information on social media and elsewhere that's an incredible burden on their mental health and I think that would be usefully understood um In The Same Spirit you know you know you don't know which um you and I don't know which News Channel to trust if we you know happen across American news channels it's very difficult to know you know is is this reliable information or not from the point of view of the free energy principle um and applying it to um understand what has gone wrong and how it could be remed um it's very very simple that you know if you if the imperatives for a sustainable maintenance of your manifold blankets and the implicit attracting set rests upon minimizing surprise and the expected surprise that underwrites your planning is literally uncertainty or ENT y the expected self- invasion is the entropy what that means is we are all in the game of choosing those actions that minimize uncertainty and coming back to what we're talking about before about the counterfactual we have to choose between different actions and if we have many many more actions available then necessarily there's going to be an entropy or uncertainty over what we're going to do and that can be incredibly that has been associated with angst so if I if I represent that and I have the hypothesis what state of mind am I in and I now don't know what to do next that is anxiety and that has direct physiological and neuroendocrine uh consequences so if that is true what that means is if I've got too many choices and I cannot trust which which uh what actions to make sorry I cannot trust um um the differential reliability of various sources of information because of this overconnectivity um that necessarily distributes the probability Mass over all the things I could do that necessarily increases my anxiety So you you're looking at the the way that we select sources of information for building our world models and our beliefs as a problem of actively selecting the right kind of data plac you know whether this is just in eye movements you know from moment to moment or whether it's actually subscribing to a particular News Channel or social group making that that is an action and it's an action exactly in the service of trying to uh maximize the expected information game that depends upon the quality of the Precision of the information if that job is made almost impossible because there are so many people shouting at me as an adolescent um um the you I can you can see immediately how one

this would lead to a a pernicious and irreducible uncertainty about how to act and select the right um the right sources of information or points of reference or Role Models um that will itself then produce um you know a me a a self moral that is imbued with anxiety and angst and just knowing not not knowing what to do next so that was the kind of pathology I was talking about right at the beginning when I was talking about overconnectivity U if you just if it's just your teacher and your mother and your father and your your priest depending upon your your culture and you in view them with an epistemic trust then this issue goes away but that does require a very sparse much more familiar communal kind of uh interaction and belief sharing and evidence accumulation is that is that the sort of um yeah we didn't get as far as talking about juwes and the legal systems use of of influence that's itself an interesting issue yeah yeah certainly thank you um I I think it's one to to sort of work through um and and take forward so really really helpful thank you then I think we are done Jonathan we are we are done yes so thank you Carl brilliant yes we brought the ship in almost a Time thank you so much um I think Carl you heard how many people were referencing you from different uh for around the place so you're certainly leaving an impression thank you so much uh could not ask for more thank you and um please keep in touch thank you all wonderful discussions thank you so much yes thank you everyone for the questions and just for listening and participating thank you good night thank you thanks Carl thanks all microphones are off but I'm sure the houses are Rippling with Applause thank you right thank you m